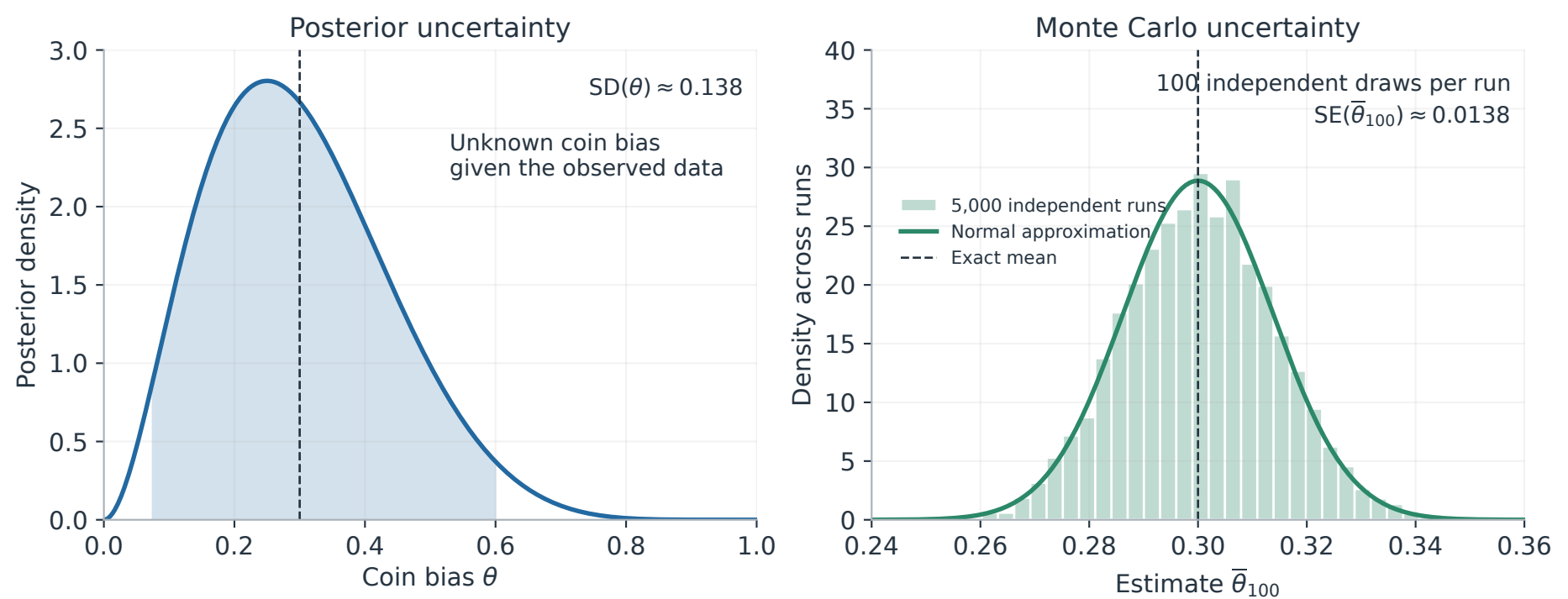
Posterior uncertainty
SD(θ) ≈ 0.138
Unknown coin bias
given the observed data
Posterior density
Coin bias θ
3.0
2.5
2.0
1.5
1.0
0.5
0.0
0.0
0.2
0.4
0.6
0.8
1.0
Monte Carlo uncertainty
100 independent draws per run
SE(θ̄100) ≈ 0.0138
5,000 independent runs
Normal approximation
Exact mean
Density across runs
Estimate θ̄100
40
35
30
25
20
15
10
5
0
0.24
0.26
0.28
0.30
0.32
0.34
0.36